

Ayman-Yereem Kone
Neural Graphics Assignment 2

Github repo: [Yereeme/A2_NeuralGraphics_AymanKone](https://github.com/Yereeme/A2_NeuralGraphics_AymanKone)

P4: Densification

Every puff gets a score, which is the average size of the gradient on its center over the 200 steps since the last pass. Every 200 steps, the top 20% by score get densified. I used a quantile because the default threshold ($2e-4$) flagged nothing here, since the scores are around $2e-5$. A flagged puff is cloned if its largest scale is at most $0.02 \times W$ (about 5 px), since a small puff just needs more coverage. Otherwise it's split into two children with scale divided by 1.6, placed randomly inside the parent. Puffs with opacity under 0.005 are pruned, and the total is capped at the budget. I reset Adam after each pass because the number of puffs changes, which the assignment clarifications allow.

I ran this on all three images, starting from 64 puffs and growing toward a 256 budget (64, 77, 93, 112, 135, 162, 195, 234, 256). Each one is compared against a plain fit with 256 puffs, using the same init and 2000 steps.

Results:

Image	Densified (final N)	Plain 256
coffee	24.91 (256)	24.80
astronaut	19.84 (255)	19.98
cat	26.16 (256)	26.20

At 256 puffs all three renders are pretty blurry. Cat is the most recognizable (both eyes and the nose are clear), while astronaut's face is mostly a soft blob, and astronaut and coffee both have a few long thin streaks that look like very stretched puffs

Original (left) vs Densified 256 (right)



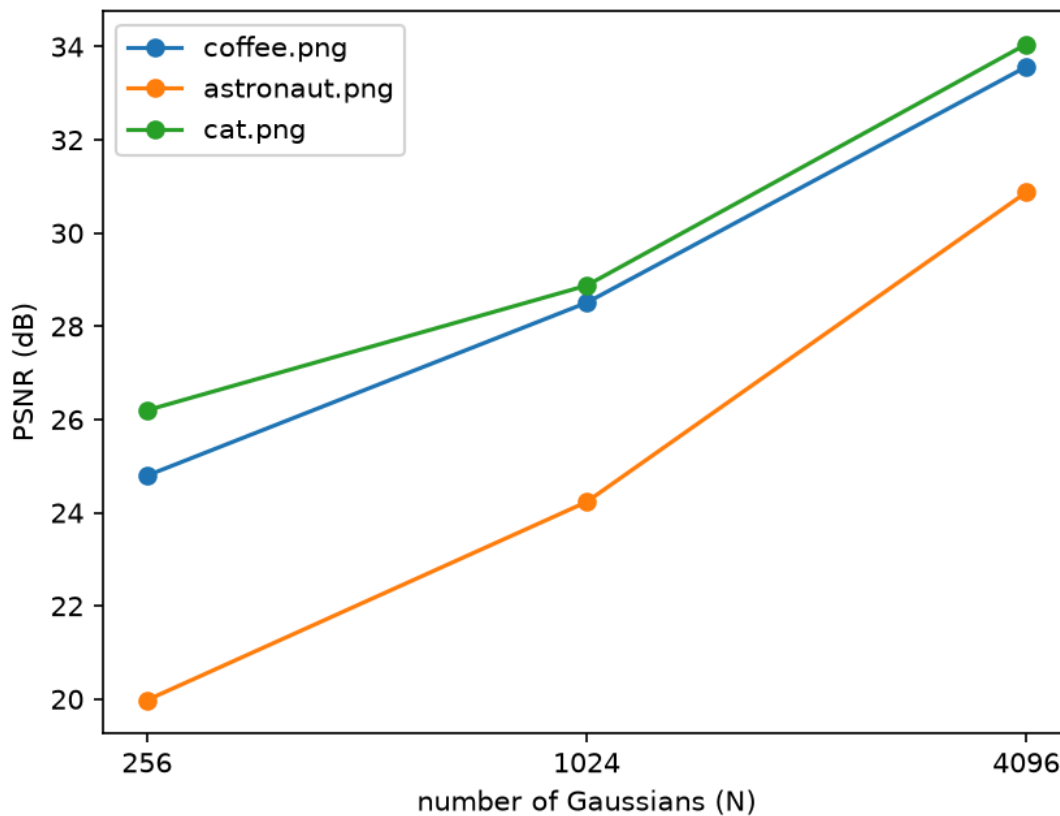
Neither one comes out ahead. Densified is higher on coffee (+0.11 dB), plain is higher on astronaut (-0.14 dB) and cat (-0.04 dB), and the averages are the same (23.64 vs 23.66 dB). These gaps are smaller than the run-to-run noise, since two runs of my final code for plain

coffee at 256 gave 24.50 and 24.80 dB from the random init alone. So I can't say densification reaches a higher PSNR here, only that it matches a plain fit while starting from a quarter of the puffs, and it also ran faster (about 34 s vs 57 s per fit). Astronaut ended at 255 instead of 256 because pruning removed one puff at the last pass, which is the only time I saw the count go down.

P5: Scaling with N

Results:

Image	N=256	N=1024	N=4096
coffee	24.80	28.51	33.57
astronaut	19.98	24.23	30.87
cat	26.20	28.87	34.05



PSNR vs number of Gaussians

4096 Gaussians renders coffee, astronaut, cat



PSNR goes up for every image every time N goes up, and none of the curves has flattened by 4096. Astronaut needs the most Gaussians. It's the lowest at 256 and 1024 (19.98 and 24.23 dB) and gains the most from 256 to 4096 (+10.9 dB, vs +8.8 for coffee and +7.9 for cat). That fits its structure, since it's the mixed portrait: big smooth areas (the suit, the backdrop) next to small detailed parts (the face, mission patch, shuttle model, flag stripes), and the detailed parts need many small puffs. Measured on the original images, astronaut has the highest contrast (pixel standard deviation 0.32, vs 0.31 for coffee and 0.17 for cat) and about 4x more fine detail than cat (the energy left after subtracting a 4x4 block average is 0.0080 vs 0.0020, and 0.0038 for coffee).

Cat is the highest at every N , which is the opposite of what I expected for dense fur. The fur is fine detail, but it's low contrast (tan on tan), so blurring it costs little PSNR. Coffee sits in the middle. It has large smooth areas (cup, saucer) that a few puffs handle well, but also sharp, high-contrast edges (cup rim, spoon, saucer against the table) that need more. The gap to cat shrinks from 1.4 dB at 256 to about 0.4 to 0.5 dB at 1024 and 4096.

At 4096 all three renders are clearly recognizable, and what's still soft is where the fine detail is: the stars and stripes on the flag and the face and patch detail on astronaut, the whiskers and

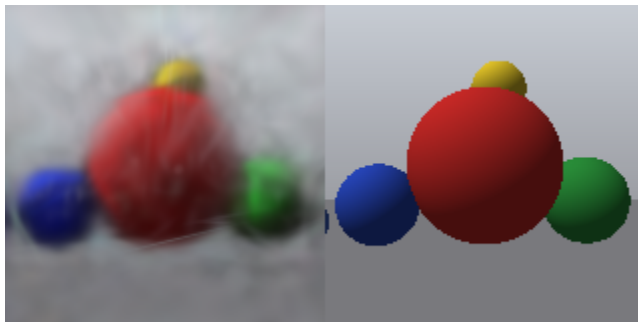
individual fur hairs on cat (the fur turns into a smooth painterly texture), and the fine wood grain in the corners of the coffee image.

P7: 3D Scene Reconstruction

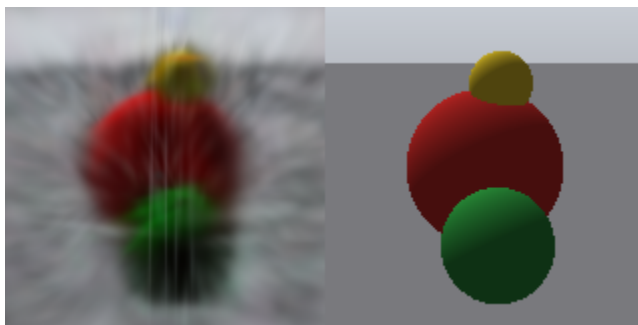
I fit 4000 Gaussians to the 44 training views (160x160) for 1500 iterations. Each step picks one random training camera, projects every Gaussian into it, sorts by camera-space depth, composites, and takes an MSE loss against that view, with Adam at lr 1e-2. For the init I used the one from the assignment: positions uniform random in the $[-1.5, 1.5]$ cube, scale 0.08 on every axis (stored in log space), identity quaternion, color 0 (so sigmoid gives gray), and opacity $\logit -2$ (about 0.12 opacity). So it starts as a coarse gray fog that the optimizer has to carve into the spheres.

Results: the mean PSNR over all 44 training views is 23.67 dB (min 20.88, max 26.54), and the fit took about 5.3 minutes. The logged PSNR was still going up at the end of the 1500 iterations, so I'd expect more iterations to help, but I stuck with the suggested budget.

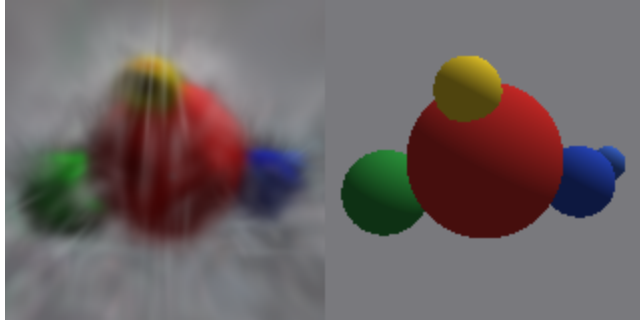
Render (left) vs ground truth (right)



Training view 0



Training view 15



Training view 30

In the renders the spheres are in the right places with the right colors, but they're pretty blurry and the edges are soft. The background is a noisy gray haze instead of a flat gray, and there are thin light streaks across and around the spheres, most visible in the second view.

P8: Densification in 3D

Going from 2D to 3D, a few things changed. Positions and scales have 3 numbers per Gaussian instead of 2. Rotation is a quaternion instead of one angle, so clone and split just copy it, and the split offset is rotated by the parent's rotation matrix built from the quaternion. The size threshold is now in world units. I used 0.1, a bit above the initial blob size of 0.08 and about 3% of the 3-unit scene. The score is still the mean size of the gradient on each center, but it's averaged over steps on randomly picked cameras between passes.

What stayed the same is basically the whole logic. The top 25% by score get densified. Small ones are cloned, big ones are split into two children at scale/1.6, anything under 0.005 opacity is pruned, and the count is capped at the budget. I rebuild Adam after every pass instead of carrying its moments over, which the instructor said was allowed.

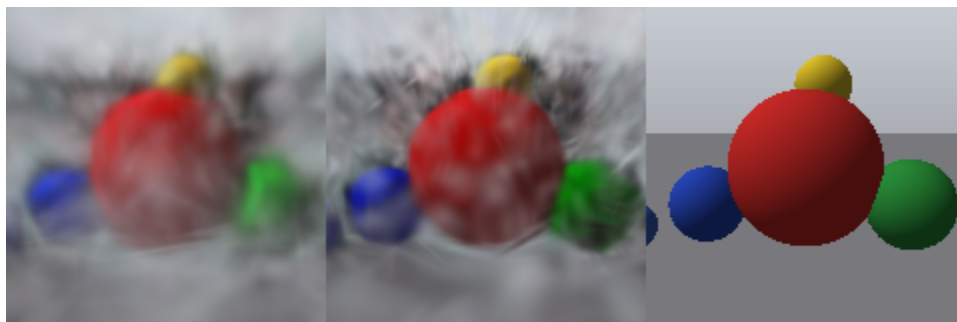
Settings: I started from 1000 Gaussians, ran a pass every 200 steps, and capped at 4000 to match the plain P7 fit, so both fits end with exactly 4000.

Results:

Fit	Gaussians	Train PSNR (mean over 44 views)	Held-out PSNR (mean over 12 views)	Worst held-out view

plain	4000	23.67	16.40	13.73
densified	4000	23.18	16.80	14.90

Plain (left), densified (middle), ground truth (right)



Held-out view 0



Held-out view 5



Held-out view 10

The densified fit is about 0.4 dB better on held-out views and 0.5 dB worse on training views. That's pretty small, and each number comes from a single run, so I wouldn't call it a clear win. Training going down a bit makes sense, since the densified fit starts from only 1000 Gaussians and has fewer steps with the full 4000. Both fits still have a gap of around 7 dB between training and held-out, so densification didn't fix the generalization problem on its own.

The clearer difference is in the opacities. The plain fit has 222 Gaussians under 5% opacity, which are nearly see-through blobs, the kind that become floaters. The densified fit has only 3. The densified fit also has far more solid Gaussians (486 above 50% opacity, against 60 for the plain fit). Nothing in either fit ended under the 0.005 prune threshold, so I can't tell from the final result how much pruning itself removed.

Looking at the three held-out views, I don't see a clear drop in haze for the densified fit. Some spheres are a bit crisper (the blue and green ones in the first view, the green one in the second), but the background has more dark streaks, and in the second view the red and yellow spheres are paler than in the plain fit. In the third view both fits are mostly gray haze with the spheres barely visible. So the opacity counts show fewer faint blobs, but that doesn't show up as an obviously cleaner picture.

P9: Novel-view orbit and held-out evaluation

For this problem I measured both fits on the 12 held-out views. I also rendered an orbit of camera positions that are not in the dataset.

Held-out numbers

Results:

	Train PSNR (mean, 44 views)	Held-out PSNR (mean, 12 views)	Gap
Plain (4000 Gaussians)	23.67	16.40	7.3 dB
Densified (1000 to 4000)	23.18	16.80	6.4 dB

The densified fit is better on held-out views by 0.4 dB. It is better on 7 of the 12 individual views. Both fits are much worse on held-out views than on training views, by 6 to 7 dB. The per-view numbers for all 12 held-out views are below.

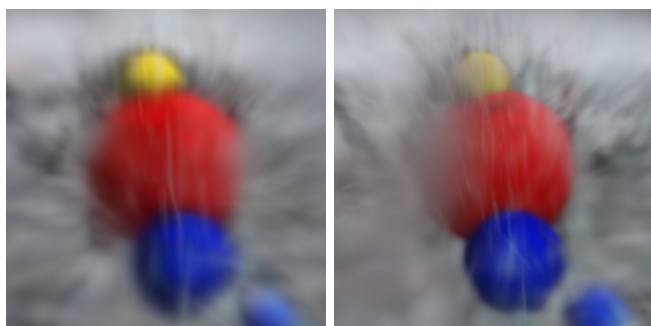
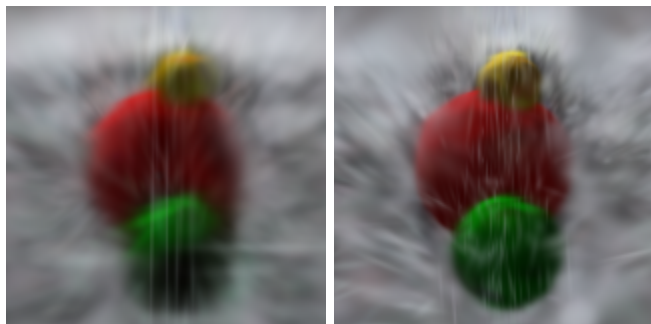
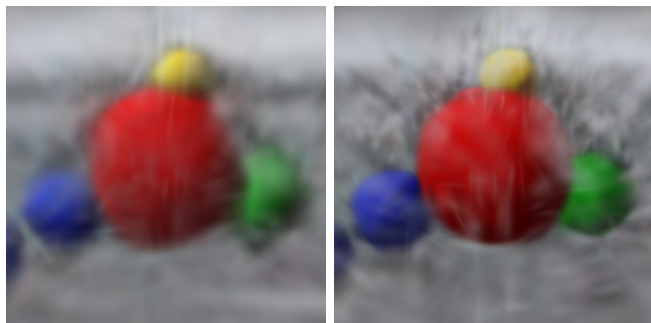
View	Plain	Densified
0	15.91	17.54
1	14.43	14.90
2	15.03	16.58
3	13.73	15.63
4	17.75	18.36
5	17.72	17.10
6	17.28	18.23
7	18.08	17.61
8	17.86	17.42
9	15.49	15.80
10	16.82	16.36
11	16.65	16.09
Mean	16.40	16.80

The worst held-out view is view 3 for the plain fit (13.73) and view 1 for the densified fit (14.90). The densified fit is clearly better on views 0, 2 and 3, and slightly worse on views 5, 7, 8, 10 and 11, so the 0.4 dB average gain is not the same everywhere.

Orbit

I also rendered 12 camera positions on a circle around the scene, none of them from the dataset. The camera is 4 units from the origin and 20 degrees above the horizon, and it looks at the origin the whole way. I rendered both fits. Plain is on the left and densified is on the right.

Orbit frames 0, 3, 6 and 9. Plain (left), densified (right)



The full 12 frames are in the submitted [p9_orbit.gif](#).

The layout stays recognizable all the way around. The red sphere stays in the middle with yellow on top. The green and blue objects change sides as the camera circles, which is what should happen. So the 3D positions are roughly right.

The renders are still blurry. The background has a gray haze with dark and white streaks, and there are fine light lines across the red sphere. In the GIF, the densified side seems to have slightly more dark streaking than the plain one. I only compared them by eye.

Where it breaks down

The biggest problem is views the fit never saw. Both fits drop 6 to 7 dB from the training views to the held-out views. The Gaussians match the 44 training views well, but between them they just guess, so held-out and orbit renders come out blurrier and noisier.

The gray haze and dark streaks in the background look like floaters. These are Gaussians that sit where nothing real is. From the training angles they are faint or hidden, so nothing pushes them away. From a new angle they show up. The low-opacity counts in P8 fit this, since many Gaussians stay faint. I did not check this directly, so it is my best explanation.

Thin structures are also hard. The fine light lines across the red sphere are an example. A 3D Gaussian is a blob, so a thin feature needs many small ones, and 4000 Gaussians is not many.

Densification only helped a little. It adds Gaussians where gradients are large, but it can't fix parts of the scene that no training view covers.

These come from single runs and I judged the images by eye, so these are my best guesses, not proven.